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Pioneering Futures Since 1898

# Part-I Recommender Systems

## Theory and Implementation

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# Learning Outcomes

By the end of this lecture on Recommender Systems, you will be able to:

- Define key concepts in Recommender Systems.
- Differentiate Collaborative Filtering, Content-Based Filtering,
- Demographics-Based Systems, and Hybrid Systems.
- Get a job in Netflix ☺

# The Whole Talk in One Slide!

Bewildered Coffee Flavours
Information Overload

How to choose my Taste?

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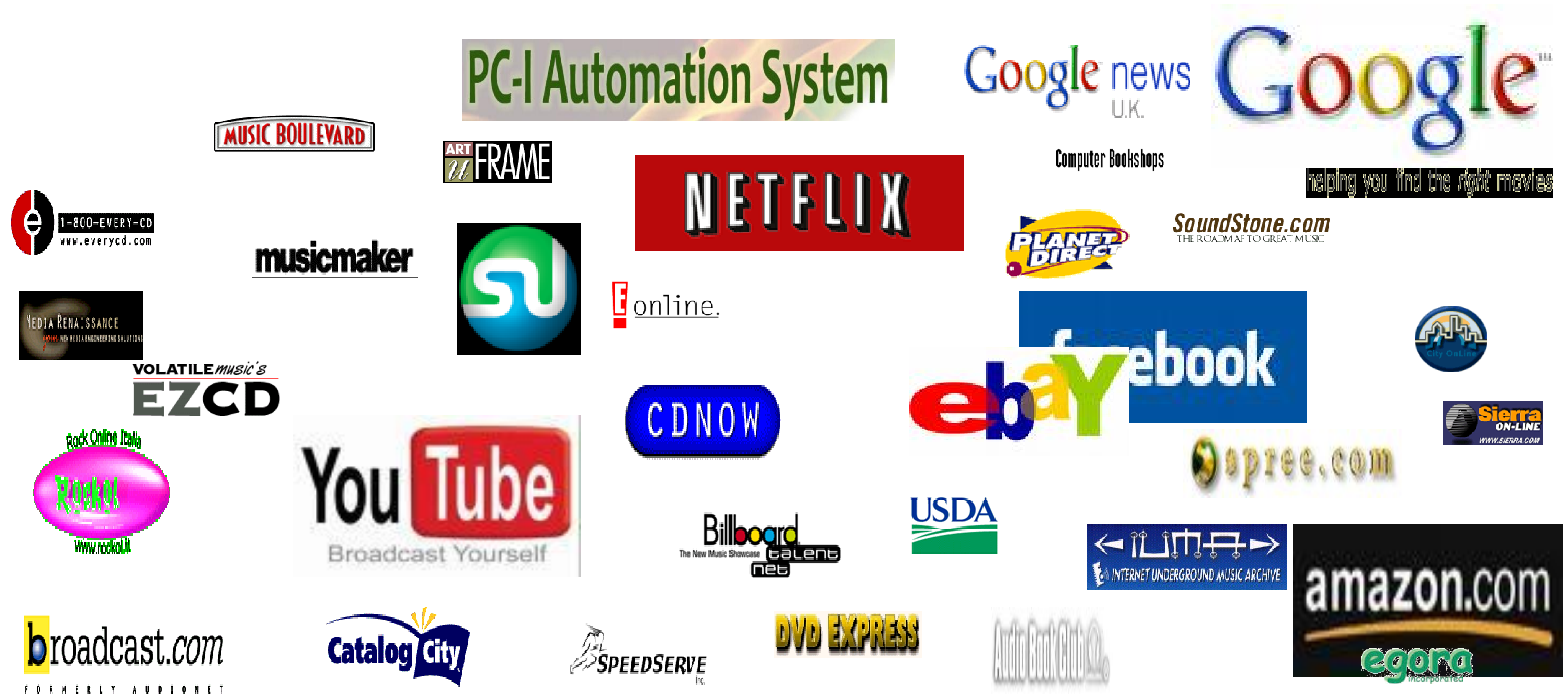
- Introduction of Recommender Systems (RS)
- Types of Recommender Systems
  - Collaborative Filtering (CF) systems
    - ✓ User based CF
    - ✓ Item Based CF
  - Content-based filtering systems (CBF)
  - Demographic-based systems (DBS)
- Concluding Remarks

# Recommender Systems (RS)

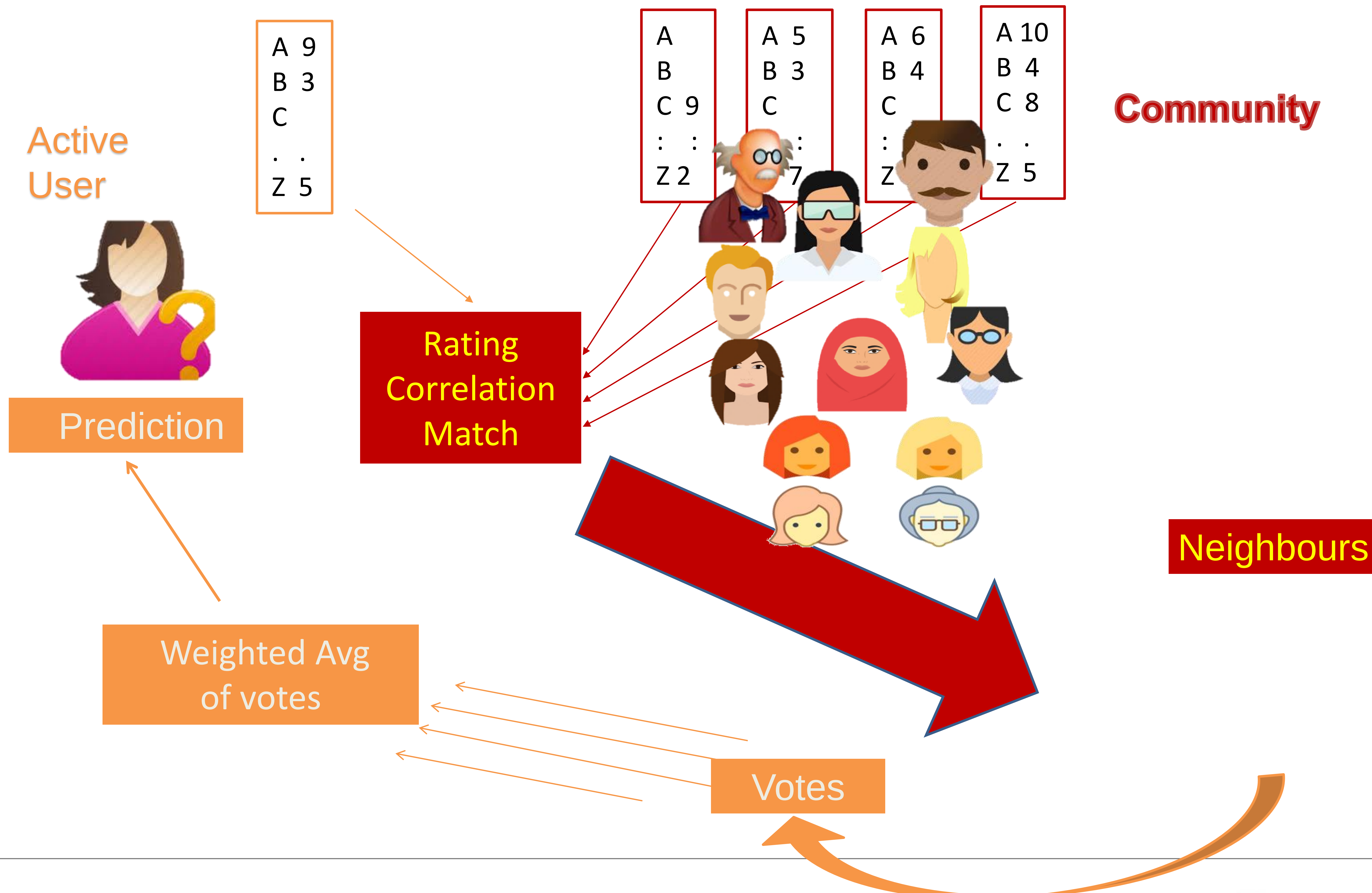
- Information filtering systems that make recommendations of items based on a model of users' preferences.
- Examples



## TOP-N Recommendation


[How Amazon Makes Business]()

# Collaborative Filtering



# Main Building Blocks

- User profile is made of ratings (about items) they provided
- Item profile is made of ratings given by other users, item’s content features
- A data base of rating exists called training set
- User-item rating matrix

## Building a Movie Recommender System--

How to predict you will like/dislike a movie prior to watch

# User-Based CF--
# User Item Rating Matrix

| | Titanic | The Proposal | XXX | Forrest Gump | The God |
|---|---|---|---|---|---|
| John | | | | | |
| Annie | | | | | |
| Ali | | | | | |
| Jollie | | | | | |

# User-Based CF

| | Titanic | The Proposal | XXX | Forrest Gump | The God |
|---|---|---|---|---|---|
| John | 10 | 10 | 8 | | x |
| Annie | 10 | 10 | 9 | | 10 |
| Ali | 1 | 2 | 1 | | 8 |
| Jollie | 1 | | 2 | | 1 |

# User-Based CF

| | Titanic | The Proposal | XXX | Forrest Gump | The God |
|---|---|---|---|---|---|
| John | 10 | 10 | 8 | | x |
| Annie | 10 | 10 | 9 | | 10 |
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Step 1: Find The Group of Users who have rated the Target Movie

# User-Based CF

| | Titanic | The Proposal | XXX | Forrest Gump | The God |
|---|---|---|---|---|---|
| John | 10 | 10 | 8 | | x |
| Annie | 10 | 10 | 9 | | 10 |
| Ali | 1 | 2 | 1 | | 8 |
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Step 2: Find The Similarity of Active User with Group of Users who have rated the Target Movie

# User-Based CF

| | Titanic | The Proposal | XXX | Forrest Gump | The God |
|---|---|---|---|---|---|
| John | 10 | 10 | 8 | | x |
| Annie | 10 | 10 | 9 | | 10 |
| Ali | 1 | 2 | 1 | | 8 |
| Jollie | 1 | | 2 | | 1 |

Step 2: Find The Similarity of Active User with Group of Users who have rated the Target Movie

# User-Based CF: Similarity Measure

$$r = \frac{\sum(X-\overline{X})(Y-\overline{Y})}{\sqrt{\sum(X-\overline{X})^2}\sqrt{(Y-\overline{Y}}}$$

Where, $\overline{X}$ = mean of X

Numpy.Corrcoef(a,b)[0,1]

Annie) = 1

```
import numpy
a=[1,2,3,0,0,0,0]
b=[2,3,4,5,10,8,5]
numpy.corrcoef(a, b)[0,1]
```

```
-0.5715430733096767
```

# User-Based CF

| | Titanic | The Proposal | XXX | Forrest Gump | The God |
|---|---|---|---|---|---|
| John | 10 | 10 | 8 | | x |
| Annie | 10 | 10 | 9 | | 10 |
| Ali | 1 | 2 | 1 | | 8 |
| Jollie | 1 | | 2 | | 1 |

Sim (John, Annie) = 1 → Very Similar

Sim (John, Annie) = 0.5 → Somewhat Similar

Sim (John, Annie) = -0.8 → Dissimilar

# User-Based CF

| | Titanic | The Proposal | XXX | Forrest Gump | The God |
|---|---|---|---|---|---|
| John | 10 | 10 | 8 | | x |
| Annie | 10 | 10 | 9 | | 10 |
| Ali | 1 | 2 | 1 | | 8 |
| Jollie | 1 | | 2 | | 1 |

Step 3: The Prediction for Target Movie for the of Active User is calculated as the SIMALARITY WEIGHTED SUM of Respective Ratings of the Neighbors



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# User-Based CF

| | Titanic | The Proposal | XXX | Forrest Gump | The God |
|---|---|---|---|---|---|
| John | 10 | 10 | 8 | | x |
| Annie | 10 | 10 | 9 | | 10 |
| Ali | 1 | 2 | 1 | | 8 |
| Jollie | 1 | | 2 | | 1 |

Prediction = {(Sim (John,Annie) * rating_Annie)
+ (Sim (John,Ali) * rating_Ali)
+ (Sim (John, Jolie) * rating_Jolie) } / Sum(|sim|)

Prediction = {(1 * 10)+ (0.5 * 8) + (-0.8 * 1) } / 2.3

Prediction = 7.5

# Item-Based CF

| | Titanic | The Proposal | XXX | Forrest Gump | The God |
|---|---|---|---|---|---|
| John | 10 | 10 | 8 | | $\boxed{x}$ |
| Annie | 10 | 10 | 9 | | 10 |
| Ali | 1 | 2 | 1 | | 8 |
| Jollie | 1 | | 2 | | 1 |

Step 1: Find The Group of Items/Movie which are similar to the Target Movie

# Item-Based CF

| | Titanic | The Proposal | XXX | Forrest Gump | The God |
|---|---|---|---|---|---|
| John | 10 | 10 | 8 | | x |
| Annie | 10 | | | | |
| Ali | 1 | | | | |
| Jollie | 1 | | | | |

Used by **Amazon** Recommender Engine

- Powerful
- Scalable
- More accurate than User-based CF

Hint: Find The Group of Items/Movie which are similar to the Target Movie

# Advantages and Disadvantages of CF

- Simple algorithm
- Use wisdom of crowd
- High quality recommendations (high accuracy)

- Accuracy suffer in sparse data
- Not very Scalable

# Content-Based Systems


- *<u>Content of items.</u>*
- Users and item profiles
- Classifiers
  - K-NN
  - SVMs,
  - Naive Bayes Classifier

Similar items scenario


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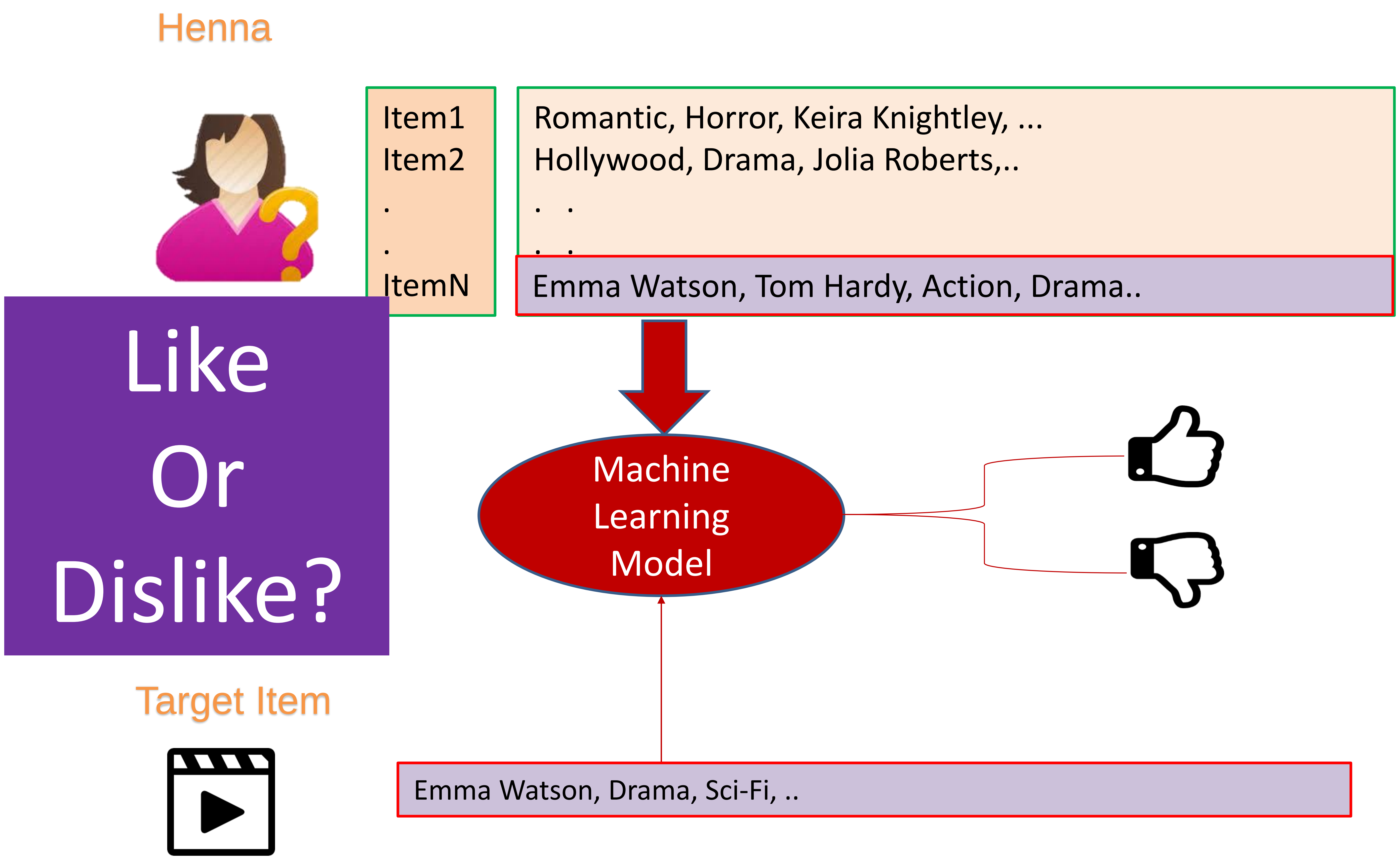
Henna
Item1
Item2
.
.
ItemN
Romantic, Horror, Keira Knightley, ...
Hollywood, Drama, Jolia Roberts,..
. .
. .
Emma Watson, Tom Hardy, Action, Drama..
Like
Or
Dislike?
Machine Learning Model
Target Item
Emma Watson, Drama, Sci-Fi, ..

# Advantages and Advantages of CBR

- Can provide explanations of recommended items by listing content-features that caused an item to be recommended.

- Requires content that can be encoded as meaningful features (problem for multimedia data).



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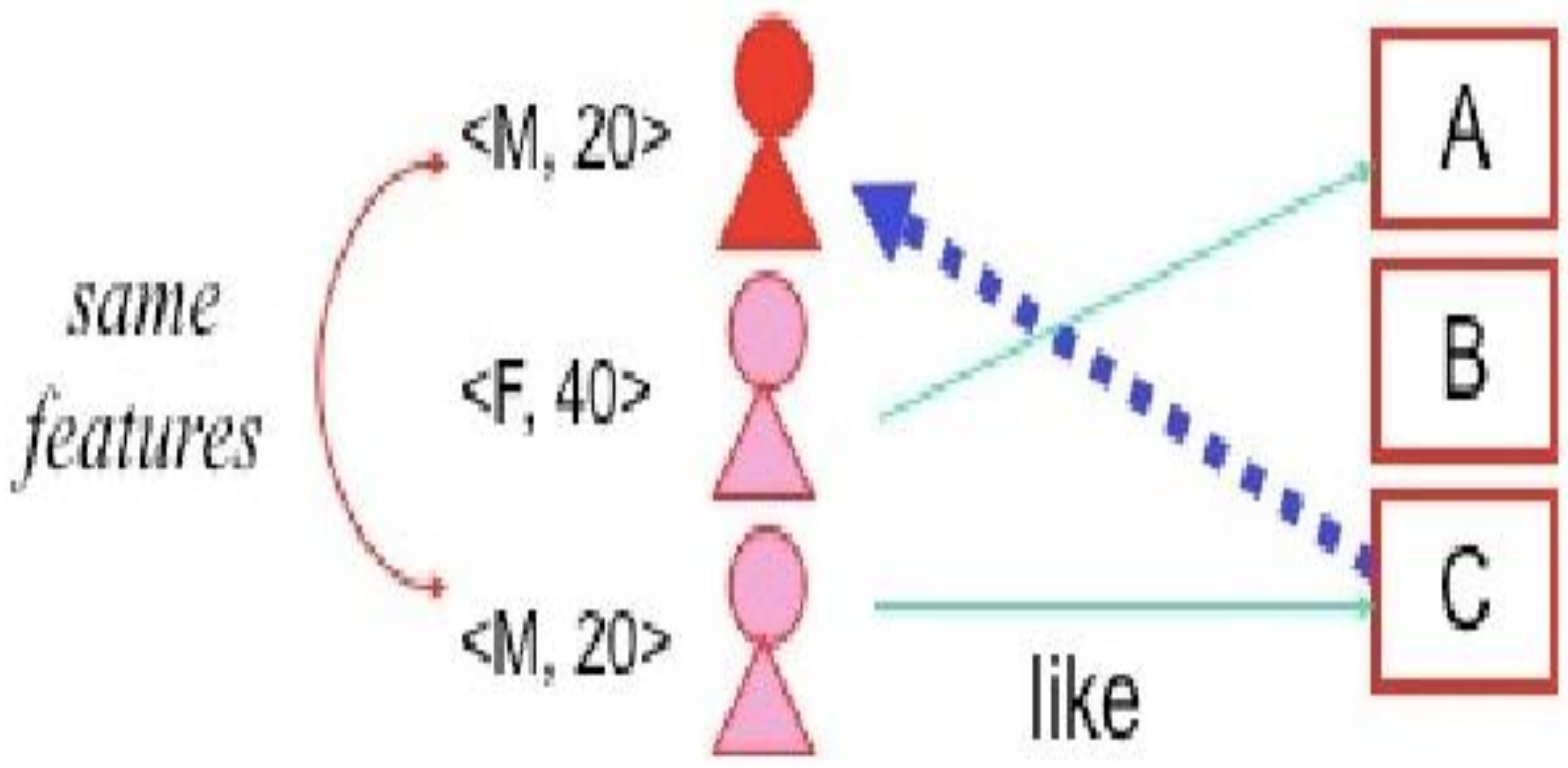
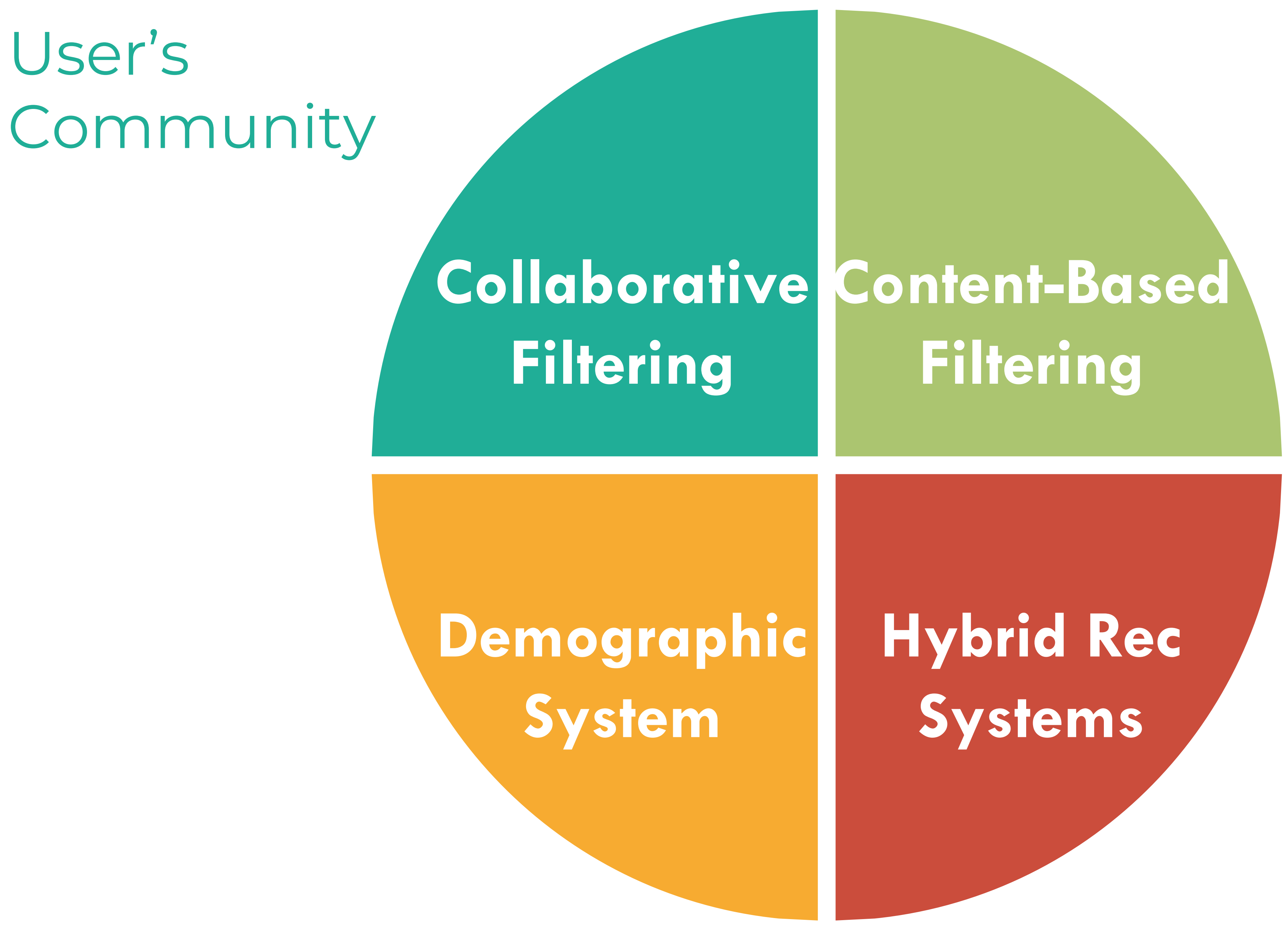
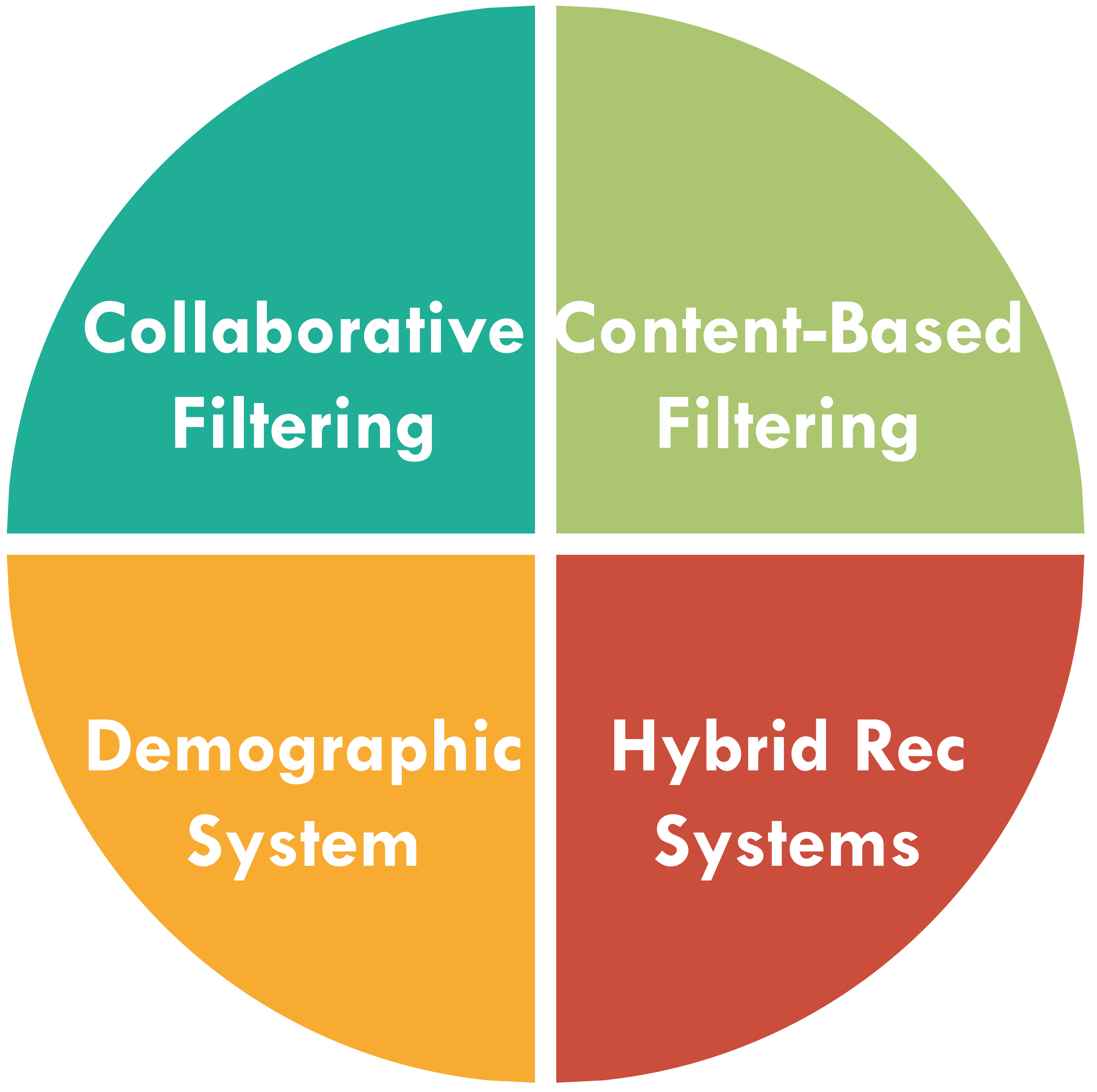
# Demographic-Based Filtering Systems

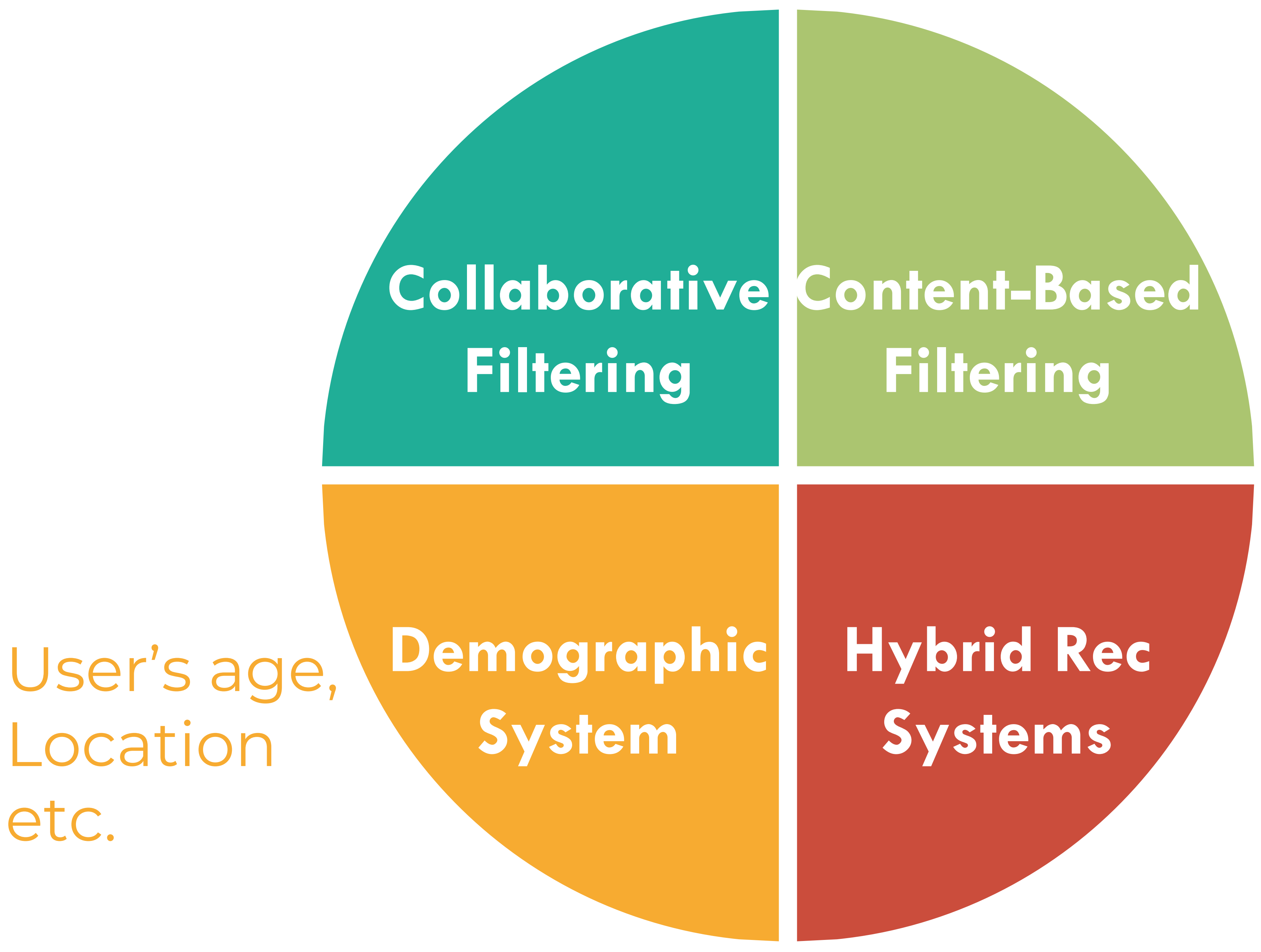



Collaborative Filtering
Content-Based Filtering
Demographic System
Hybrid Rec Systems

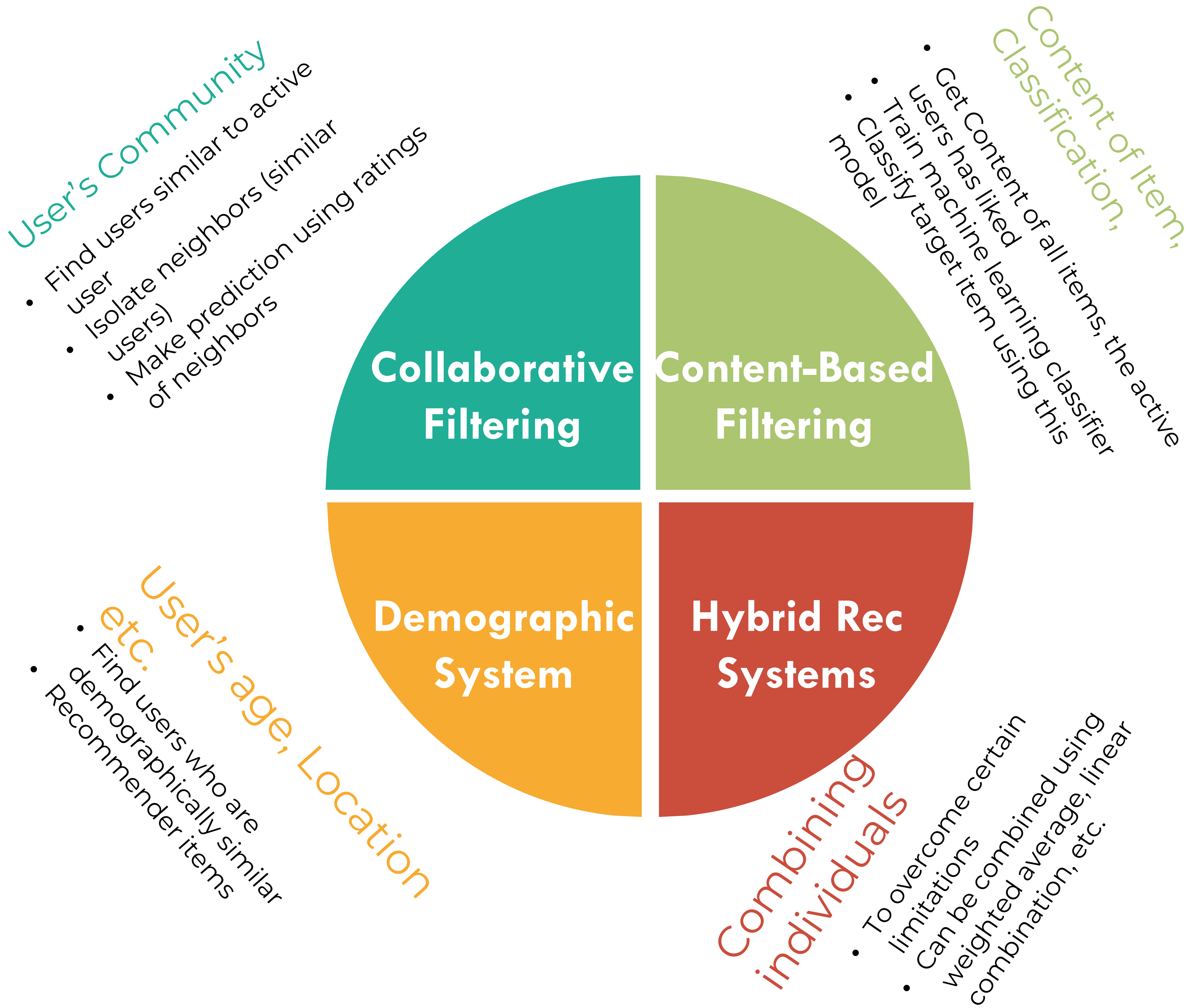

User's Community
Collaborative Filtering
Content-Based Filtering
Demographic System
Hybrid Rec Systems




Content of
Item,
Classification,

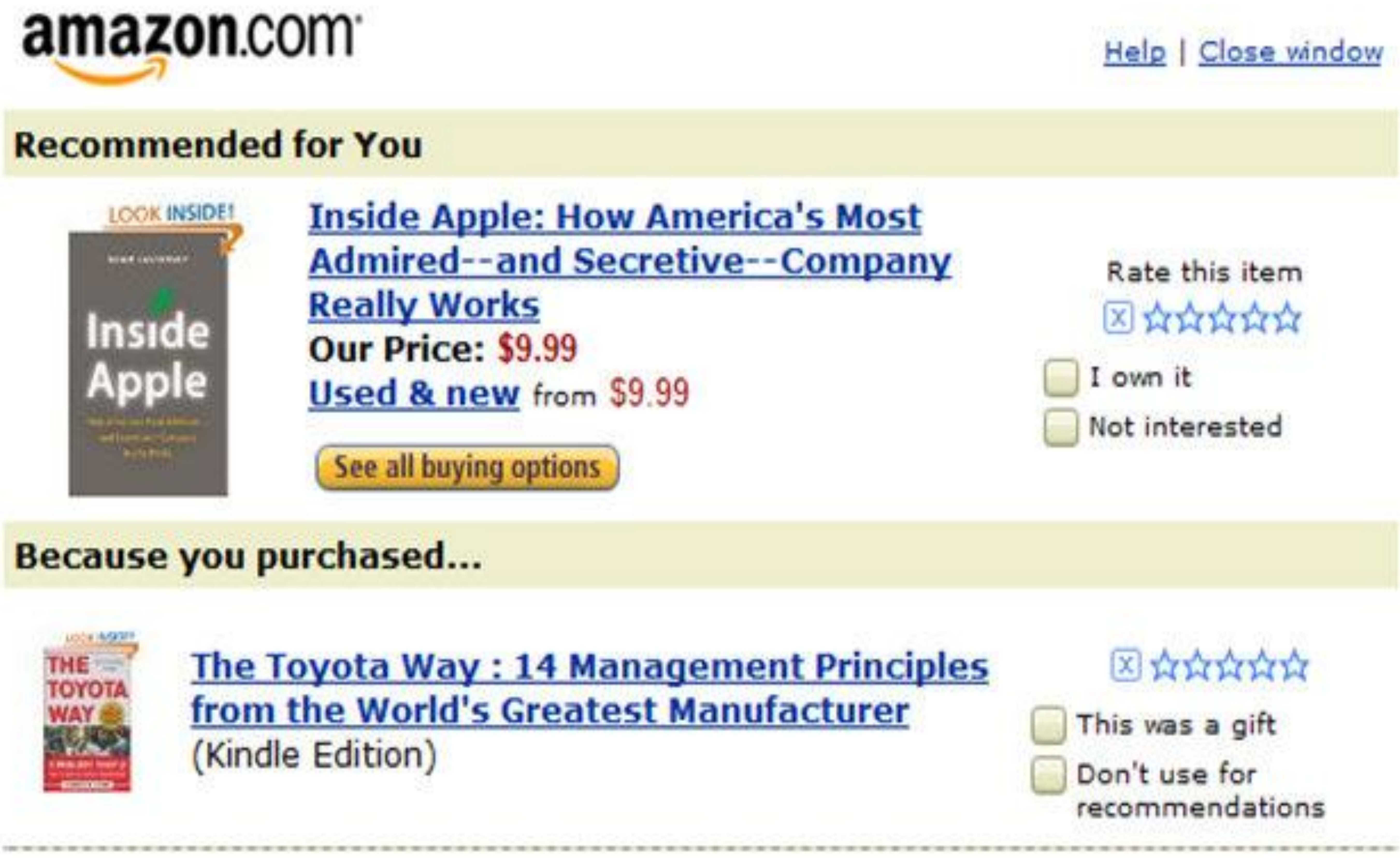

Collaborative Filtering
Content-Based Filtering
Demographic System
Hybrid Rec Systems
User's age, Location etc.


Combining individuals



Collaborative Filtering
Content-Based Filtering
Demographic System
Hybrid Rec Systems
User's Community
Find users similar to active user
Isolate neighbors (similar users)
Make prediction using ratings of neighbors
Content of Item, Classification,
Get Content of all items, the active users has liked
Train machine learning classifier
Classify target item using this model
User's age, Location etc.
Find users who are demographically similar
Recommender items
Combining individuals
To overcome certain limitations
Can be combined using weighted average, linear combination, etc.


PATTERN RECOGNITION
AND MACHINE LEARNING
CHRISTOPHER M. BISHOP
DATA
MINING
Collective
Intelligence
in Action




[How Amazon Makes Business]

# References and Online Resources

- LinkedIn: https://www.linkedin.com/learning/comptia-data-plus-da0-001-cert-prep-domain-3-0-data-analysis/data-analysis?u=56744289
- Interview Tips: https://www.linkedin.com/advice/1/what-best-ways-prepare-data-analysis-interviews
- Credit: The slides have been adapted from Dr. Amin Karimi (UEL)